

ALGEBRA 1 – MAT 1207

TUTORIAL 1

1. Use truth table to show that the following statements are logically equivalent.

- a) $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$
- b) $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$
- c) $P \Rightarrow Q = (\sim P) \vee Q$
- d) $\sim(P \vee Q) = (\sim P) \wedge (\sim Q)$
- e) $\sim(P \vee Q \vee R) = (\sim P) \wedge (\sim Q) \wedge (\sim R)$
- f) $\sim(P \wedge Q \wedge R) = (\sim P) \vee (\sim Q) \vee (\sim R)$
- g) $P \Rightarrow Q = (P \wedge \sim Q) \Rightarrow (Q \wedge \sim Q)$
- h) $\sim P \Leftrightarrow Q = (P \Rightarrow \sim Q) \wedge (\sim Q \Rightarrow P)$

2. Decide whether or not the following pairs of statements are logically equivalent.

- a) $P \wedge Q$ and $\sim(\sim P \vee \sim Q)$
- b) $(P \Rightarrow Q) \vee R$ and $\sim((P \wedge \sim Q) \wedge \sim R)$
- c) $(\sim P) \wedge (P \Rightarrow Q)$ and $\sim(Q \Rightarrow P)$
- d) $\sim(P \Rightarrow Q)$ and $P \wedge \sim Q$
- e) $P \vee (Q \wedge R)$ and $(P \vee Q) \wedge R$
- f) $P \wedge (Q \vee \sim Q)$ and $(\sim P) \Rightarrow (Q \wedge \sim Q)$

3. (a). Draw Venn diagrams to illustrate each of the following identities:

- I) $A - B = A \cap B'$
- II) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- III) $(A \cap B)' = A' \cup B'$
- IV) $(A - B) \cap C = (A \cap C) - B$

(b). Given that $|A| = 55$, $|B| = 40$, $|C| = 80$, $|A \cap B| = 20$, $|A \cap B \cap C| = 17$, $|B \cap C| = 24$ and $|A \cup C| = 100$. Find,

- I) $|A \cap C|$,
- II) $|(B \cup C) - (A \cap B \cap C)|$,
- III) $|(A \cup B \cup C)'|$, if $|\mathcal{E}| = 150$.

4. For any two subsets A and B of a given set S define $A - B$, the relative complement of B with respect to A. Using the definition of equality of two sets show that,

- I) $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$,
- II) $(A - B) - C = A - (B \cup C) = (A - B) \cap (A - C)$,
- III) $A - (B \cap C) = (A - B) \cup (A - C)$.

5. Let A, B and C be subsets of a given set S.

a) Prove that

- I. if $A \cap B = \emptyset$ then $A \subseteq B'$,
- II. $A' - B' = B - A$,
- III. $A \subseteq B \Rightarrow A \cup (B - A) = B$,
- IV. $A \cap (B - C) = (A \cap B) - (A \cap C)$

b) Let the operator Δ be defined by $A \Delta B = (A - B) \cup (B - A)$. Prove that,

- I. $A \Delta (B \Delta C) = (A \Delta B) \Delta C$,
- II. $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$,
- III. $A \Delta B = \emptyset$ if and only if $A = B$.

6. Let A, B and C be subsets of a given set S. Using the Laws of Algebra of sets simplify the following expressions:

- I. $(A \cup B) \cup (B' \cap A) \cup (A' \cap B)'$
- II. $A \cup (A \cap B)' \cup [B' \cap (A' \cup C)] \cup (C \cap A')$
- III. $[A' \cap (B \cup C)] \cup [B' \cap (A \cup C)] \cup [C' \cap B \cap A']$

7. Consider the following relations on the set of integers.

- I. $R_1 = \{(a, b) | a \leq b\}$
- II. $R_2 = \{(a, b) | a > b\}$
- III. $R_3 = \{(a, b) | a = b \text{ or } a = -b\}$
- IV. $R_4 = \{(a, b) | a + b \leq 3\}$

Which of these relations are reflexive, symmetric, anti-symmetric, or transitive?

8. Consider the following relations on $\{1, 2, 3, 4\}$:

- I. $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$
- II. $R_2 = \{(1,1), (1,2), (2,1)\}$
- III. $R_3 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$
- IV. $R_4 = \{(3,4)\}$

Which of these relations are reflexive, symmetric, anti-symmetric, or transitive?

9. Let R_1 and R_2 be equivalence relations on a set S . show that the relation R on S , defined by xRy if and only if xR_1y and xR_2y , is an equivalence relation. Is the relation $\sim$ on S , defined by $x \sim y$ if and only if xR_1y or xR_2y , an equivalence relation?

10. (a). A relation R on $S = \{(x, y) | x, y \in \mathbb{Z}\}$ is defined by $(a, b)R(c, d)$ if $a^2 + d^2 = c^2 + b^2$. Is R an equivalence relation? Justify your answer.

(b). Show that relation R defined on the set of real numbers $\mathcal{R}$ by xRy if $x^2y - y^2x - 2(x - y) = 0$ is an equivalence relation on $\mathcal{R}$.

11. A relation on R on a set S satisfies the following two properties:

I. R is reflexive.

II. For all $a, b, c \in S$. aRb and bRc imply cRa .

Prove that R is an equivalence relation.